



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO YUGA AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

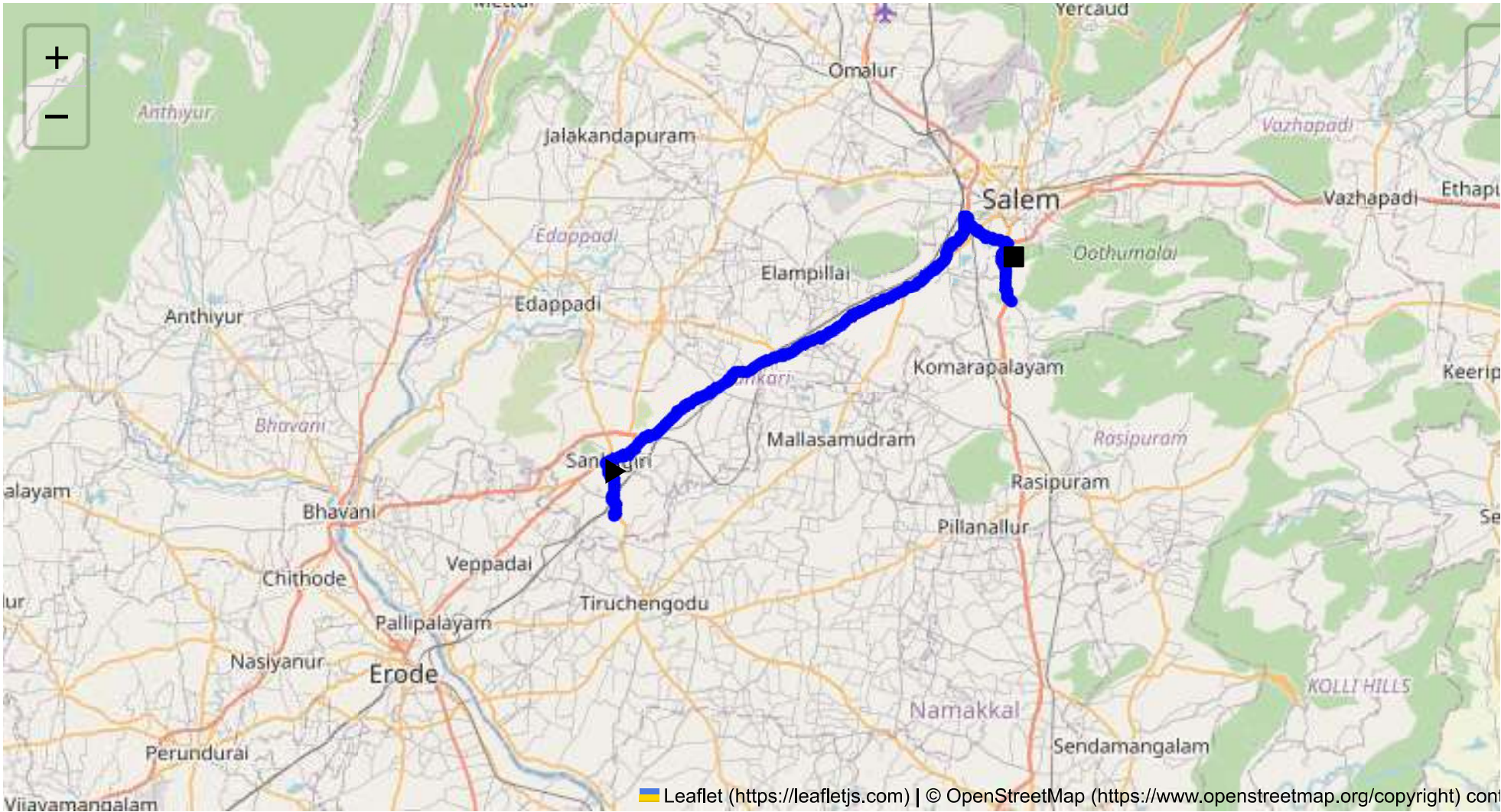
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 47.93 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.58487, 78.15167)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 4B, Gajjalnaickenpatti, Salem stretches approximately 47.93 kilometers, primarily following NH 544 and NH 79. The route is straightforward with several junctions connecting smaller towns and rural areas to the main highway. As a key corridor in Tamil Nadu, it supports considerable traffic, including heavy vehicles transporting goods.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences three main seasons: summer (March to June), monsoon (July to September), and winter (October to February). During the monsoon season, heavy rainfall can lead to waterlogging and reduced visibility, increasing the risk of accidents. The summer months might create heat-induced road stress, potentially affecting vehicle performance and road integrity.

3. Traffic Patterns and Congestion-Prone Areas

Traffic tends to peak during morning (7:00 - 9:00 AM) and evening (5:00 - 8:00 PM) hours as local populations commute to work. Key congestion points include the intersections near Sankari and Salem, where traffic flow from different towns converges. Salem, being a significant city, can experience heavy traffic, especially around market areas and transportation hubs.

4. Assessment of Road Quality and Infrastructure

The NH 544 is generally well-maintained, with clear signage, marked lanes, and relatively smooth road surfaces. However, the transition to smaller roads with reduced lane width near villages might require caution due to potential potholes and less frequent maintenance.

5. Suggestions for Alternative Routes for Emergencies

In case of roadblocks or significant delays on NH 544, utilizing SH 88 or SH 86 can be viable alternatives. However, these routes may have varying infrastructure quality and can pass through more densely populated regions, potentially affecting travel times.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to national safety guidelines, including proper labeling, use of protective gear, and route-specific permissions. Nighttime travel with hazardous materials may be restricted or need additional approvals.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

While specific data on incidents is limited, NH 544 has witnessed accidents involving heavy vehicles, often attributable to high speeds or equipment failure. Local news archives indicate occasional severe weather has exacerbated certain incidents.

8. Environmental Considerations and Sensitive Areas

The route passes near agricultural lands and small water bodies, which may be environmentally sensitive. Precautions must be taken to prevent spillage or contamination, especially when transporting hazardous materials.

9. Analysis of Communication Coverage

Most of the route is covered by major telecommunications providers, ensuring reliable communication. However, minor dead zones may occur in less densely populated areas or where the terrain is elevated.

10. Estimated Emergency Response Times for Different Route Segments

Response times within urban and semi-urban areas along NH 544 are usually around 20-30 minutes due to nearby facilities. In rural stretches or during peak weather conditions, times could extend to 45 minutes to 1 hour.

11. Overall Summary of Risk Assessment

While the primary route from Sangagiri to Gajjalnaickenpatti is generally safe for heavy vehicle travel, several potential risks require attention:

- Weather Hazards:** Monsoon rains pose the greatest risk for reduced visibility and control.
- Traffic Congestion:** Peak hours congestion can delay travel, particularly near Salem.
- Road Conditions:** Good overall, but caution is advised on minor roads and during severe weather.
- Regulatory Compliance:** Strict adherence is necessary for transporting hazardous materials, especially concerning timings and clearances.
- Environmental Impact:** Care must be taken to prevent environmental contamination.

Drivers should be trained to handle unexpected situations, equipped with communication tools, and provided with knowledge of alternative routes and emergency contacts.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	High	11.45004, 77.87206	15 KM/Hr
6	Turn	Medium	11.45028, 77.87202	30 KM/Hr
7	Turn	High	11.47403, 77.86841	15 KM/Hr
8	Turn	Medium	11.47410, 77.86845	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.49149, 77.89601	15 KM/Hr
10	Turn	High	11.49224, 77.89601	15 KM/Hr
11	Turn	Medium	11.64156, 78.12162	30 KM/Hr
12	Turn	Medium	11.64116, 78.12168	30 KM/Hr
13	Turn	Medium	11.62509, 78.14758	30 KM/Hr
14	Blind Spot	Blind Spot	11.58757, 78.14819	10 KM/Hr
15	Turn	High	11.58771, 78.14873	15 KM/Hr
16	Blind Spot	Blind Spot	11.58463, 78.15140	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
24	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
25	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
26	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
27	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
28	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
29	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
30	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
31	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
32	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
33	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
34	pharmacy	மருந்தகம்	11.6187131, 78.1451978	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium